PUDI SAI KIRAN

GCP Data Engineer — Generative AI Engineer — Agentic AI Engineer

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Professional Summary

GCP Data Engineer and Generative AI Engineer with 2 years of professional experience building cloud-native data engineering, Generative AI, and Agentic AI solutions. Hands-on experience with Google Cloud Platform services including BigQuery, Cloud Storage (GCS), Cloud Composer (Apache Airflow), Dataflow, Cloud Run, Cloud Functions, Secret Manager, and Vertex AI.

Experienced in designing and orchestrating ETL/data pipelines using Python, SQL, Apache Airflow, Cloud Composer, BigQuery, GCS, and Dataflow. Strong experience in Agentic AI systems using Vertex AI, Gemini models, LangGraph, MCP, A2A patterns, RAG architectures, tool-augmented agents, grounding, structured prompting, and multi-agent orchestration.

Proficient in developing production-grade AI and data applications using Python, FastAPI, LangChain, LangGraph, vector databases, REST APIs, Docker, and CI/CD. Experienced in integrating enterprise data pipelines with AI-driven monitoring, root cause analysis, automation, governance, and self-healing workflows.

Technical Skills

Programming: Python, SQL, HTML/CSS
GCP Data Engineering: Google Cloud Platform (GCP), BigQuery, Cloud Storage (GCS), Cloud Composer, Apache Airflow, Dataflow, Cloud Run, Cloud Functions, Secret Manager, Cloud Scheduler
Data Engineering & ETL: ETL/ELT Pipelines, Data Pipelines, Data Transformation, Data Orchestration, Workflow Automation, Data Processing, Batch Processing, Pipeline Monitoring, Data Quality
BigQuery: SQL, Data Warehousing, Query Optimization, Data Transformation, Analytical Queries, Dataset/Table Management, Reporting Pipelines
Airflow & Orchestration: Apache Airflow, Cloud Composer, DAG Development, Task Dependencies, Scheduling, Pipeline Monitoring, Retry Handling, Error Handling, Workflow Orchestration
Generative & Agentic AI: Vertex AI, Gemini Models, Vertex AI Studio, MCP (Model Context Protocol), Tool-Augmented Agents, Multi-Agent Systems, RAG, Grounding, Embeddings, Prompt Engineering, Structured Prompting, Context Memory, Safety Layers
AI Frameworks & APIs: LangChain, LangGraph, LangSmith, FastAPI, Flask, HuggingFace Transformers, OpenAI API, Vertex AI SDK
Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Semantic Similarity
Deep Learning: Neural Networks, CNN, RNN, LSTM, Attention Models, Transfer Learning
Vector Databases: ChromaDB, Pinecone, FAISS, Weaviate, PostgreSQL + pgVector
Databases: PostgreSQL, MySQL, MongoDB
Cloud & Deployment: Docker, Cloud Run, AWS EC2, AWS ECR, GitHub Actions, CI/CD
Security & Secrets: Google Cloud Secret Manager, Token-Based Authentication, Role-Based Access Control
Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Postman, Power BI

Professional Experience

Generative AI Developer / GCP Data Engineer

- Developed and deployed Agentic AI solutions on Google Cloud Platform using Vertex AI and Gemini models, implementing tool-augmented agents, A2A coordination patterns, and multi-agent workflows for enterprise automation.
- Designed and maintained cloud-native data pipelines using GCP Dataflow, BigQuery, Cloud Storage (GCS), and Cloud Composer (Apache Airflow) for structured and unstructured enterprise data processing.
- Developed Apache Airflow DAGs in Cloud Composer for ETL orchestration, task dependencies, scheduling, retry handling, failure management, and automated pipeline execution.
- Built BigQuery-based data transformation and analytics workflows using SQL, enabling scalable processing and reporting across enterprise datasets.
- Implemented secure credential and configuration management using Google Cloud Secret Manager, reducing dependency on hard-coded credentials and improving application security.
- Built conversational AI applications handling 500+ user queries/day using Gemini, structured prompting, grounding, and safety layers, improving response accuracy and engagement by 30%.
- Designed RAG-based retrieval systems using vector embeddings and semantic search, reducing response latency by 40%.
- Built multi-agent architectures using LangGraph, MCP, and A2A patterns to automate complex business workflows and intelligent decision-making processes.
- Developed AI-powered pipeline monitoring and root cause analysis workflows for identifying Dataflow failures, schema issues, infrastructure problems, and data-quality violations.

- Automated failed pipeline recovery and remediation workflows with adaptive retry logic, resource scaling, and approval-based execution.
- Developed scalable FastAPI/Flask microservices for AI and data applications, integrating REST APIs with cloud-native services and enterprise workflows.
- Deployed containerized applications using Docker on GCP Cloud Run and AWS EC2/ECR with CI/CD pipelines using GitHub Actions.
- Built data processing pipelines for PDFs, web data, and structured datasets using Python, GCP Dataflow, BigQuery, and cloud storage.

Projects

AI-Powered Pipeline Monitoring & Self-Healing System (Unilever Client) LangGraph, FastAPI, GCP Dataflow, BigQuery, Cloud Composer

- Developed an enterprise-grade Agentic AI system for automated pipeline monitoring, intelligent root cause analysis, and self-healing orchestration across GCP data processing environments.
- Designed LangGraph-based multi-agent orchestration workflows containing Observer, RCA, Decision, Governance, Data Quality, Escalation, and Self-Healing agents.
- Integrated GCP Dataflow and BigQuery workflows for monitoring pipeline execution, identifying failures, analyzing data-processing issues, and generating operational insights.
- Developed AI-driven RCA (Root Cause Analysis) workflows using LLMs and historical incident retrieval to identify pipeline failures, schema drift, infrastructure issues, and policy violations.
- Implemented a Decision Agent capable of routing incidents to self-healing, governance, escalation, or data-quality workflows based on confidence scoring and business impact.
- Developed automated self-healing workflows to retrigger failed GCP Dataflow pipelines using adaptive retry logic, resource scaling, and approval-based execution mechanisms.
- Integrated human-in-the-loop approval workflows using LangGraph interrupt mechanisms for controlled execution of high-impact remediation actions.
- Built governance and data-quality agents for schema validation, compliance checks, PII detection, and anomaly assessment within enterprise data pipelines.
- Implemented real-time monitoring dashboards and WebSocket-based workflow tracking for live agent execution visibility and operational insights.
- Developed a knowledge-base-driven incident learning system using ChromaDB to store and retrieve historical RCA patterns and remediation decisions.

ATS Resume Job Matching & AI Recruiter Assistant Platform FastAPI, Azure OpenAI, LangGraph, ChromaDB, MCP

- Developed an enterprise ATS platform for intelligent resume-job description matching using AI embeddings and semantic similarity search.
- Built end-to-end resume and job-description processing pipelines including PDF parsing, skill extraction, candidate ranking, and automated qualification analysis.
- Implemented an AI-powered recruiter search system allowing recruiters to query candidate profiles using natural language prompts.
- Developed semantic candidate retrieval workflows using embeddings, TF-IDF ranking, semantic search, and LLM-based intent detection.
- Built resume segregation workflows with automated match-percentage scoring and qualification analysis.
- Integrated Azure OpenAI models for recruiter summaries, missing-keyword analysis, and AI-generated candidate evaluation insights.
- Developed scalable FastAPI APIs supporting concurrent resume evaluation and enterprise recruitment workflows.

AI Support Automation & MCP Agent Platform (Bloon Client) FastMCP, Azure OpenAI, HubSpot, FastAPI

- Developed an enterprise AI support automation platform using MCP and Agentic AI workflows for intelligent ticket management and customer support automation.
- Built a multi-tool AI assistant supporting document search, ticket creation, ticket updates, ticket escalation, and previous-issue resolution using HubSpot APIs.
- Implemented intelligent query routing using AI-based intent detection to dynamically select document retrieval, ticket operations, and conversational workflows.
- Developed semantic previous-ticket retrieval to identify similar historical issues and generate automated resolution suggestions.
- Built AI-powered ticket lifecycle workflows covering ticket creation, status tracking, escalation, priority management, and response summarization.
- Integrated Azure OpenAI models for conversational AI, issue classification, workflow reasoning, and customer interaction automation.
- Developed scalable asynchronous FastAPI and MCP server architecture with modular AI tools, concurrent request handling, and logging pipelines.

BCG AI Analytics & MCP Intelligence Platform (BCG Client) Python, MCP, FastAPI, Azure OpenAI

- Developed an enterprise MCP server for natural-language analytics and database intelligence across multiple operational datasets.
- Built an AI-powered SQL generation framework capable of converting business questions into secure read-only SQL queries using LLM orchestration and schema-aware prompting.
- Designed table-specific MCP tools and schema-driven configurations enabling scalable analytics across enterprise databases.
- Implemented hybrid analytics workflows combining database intelligence with Jira ticketing integration using run_id-based correlation.
- Developed intelligent SQL agent workflows using semantic KPI retrieval, contextual prompt engineering, and automated answer generation.
- Built database introspection tools supporting schema discovery, table definitions, metadata exploration, and automated analytics generation.
- Engineered secure query validation to prevent INSERT, UPDATE, DELETE, DROP, and other write operations.
- Developed natural-language analytics workflows supporting trend analysis, governance monitoring, pipeline insights, incident intelligence, and operational reporting.
- Implemented automated Excel reporting and downloadable business intelligence reports.

HR Self-Service AI Assistant (RAG-Based Enterprise Chatbot) FastAPI, ChromaDB, Azure OpenAI, RAG

- Developed an enterprise HR self-service AI chatbot enabling employees to query HR policies, leave rules, reimbursement guidelines, onboarding procedures, and internal documentation.
- Built an end-to-end RAG pipeline for document ingestion, chunking, embedding generation, vector indexing, retrieval, and contextual response generation.
- Implemented document ingestion workflows supporting PDF uploads, metadata extraction, automated chunking, and vector database indexing.
- Designed semantic search architecture using embeddings and ChromaDB for contextual retrieval across enterprise knowledge repositories.
- Developed FastAPI backend APIs for document upload, chat interaction, authentication, session management, and conversational history.
- Built asynchronous LLM orchestration workflows with concurrency optimization, timeout handling, and scalable chatbot processing.
- Implemented token-based authentication and role-based document access controls for multi-user enterprise environments.
- Developed conversational workflows supporting contextual follow-up questions, chat memory, feedback collection, and personalized assistance.

Achievements

- Solved 100+ problems on LeetCode and 50+ problems on GeeksforGeeks.
- Built and deployed multiple end-to-end Agentic AI applications using Vertex AI, Gemini models, LangGraph, MCP, vector databases, and GCP cloud-native services.
- Hands-on experience building GCP data pipelines using BigQuery, Cloud Composer, Apache Airflow, Dataflow, Cloud Storage, and Secret Manager.
- Actively exploring Generative AI advancements including Vertex AI Agent Builder, Gemini multimodal capabilities, and JAX-based model optimization.

Education

MVGR College of Engineering, Vizianagaram 2024
B.Tech Electrical & Electronics Engineering (Major), Computer Science (Minor)
CGPA: 8.21 (EEE), 8.0 (CSE)

Certifications

- Salesforce Developer Internship – Smart Internz
- NPTEL Joy of Computing using Python
- Google Cloud Platform – Fundamentals of Cloud Computing (In Progress)